



COURSE 420-D01-AB · COHORT CYB-01 · DAYS 05 & 06

Installing and Configuring Monitoring Tools

4 hours lecture · 4 hours hands-on · Competency EZ01 element 3

Lab 3 and Lab Environment write-up feed the Course Procedure Manual.

From the Industry to the Classroom — Building the IT's Future, Today and Together!



The four jobs of monitoring

- ▶ **Packets** — what bytes crossed the wire? (Wireshark)
- ▶ **Metrics** — how busy/healthy is the device? (Cacti, SNMP, Prometheus)
- ▶ **Logs** — what did the device just tell us? (syslog, Splunk, ELK)
- ▶ **Alerts** — which of the above demands a human? (Nagios, Wazuh)

Key takeaway — *Packets, metrics, logs and alerts are four different food groups. A healthy monitoring diet has some of all four.*



By the end of Day 06 you can...

- ▶ Explain the role and history of the major free monitoring tools
- ▶ Build a reproducible lab on **VirtualBox + GNS3 + AWS Academy Learner Lab**
- ▶ Install Wireshark, SNMP, Wazuh and Splunk on Windows and Linux
- ▶ Configure baseline alerts through pfSense or OPNsense
- ▶ Document everything as a screenshot-rich procedure manual

Key takeaway — *Five outcomes, two days. Every outcome is graded somewhere in the Course Procedure Manual.*



SECTION A

Day 05 — Lecture 1

Discovery, Polling, Logging



Packet analysis — the standard

History and foundations

- ▶ Created **1998** by Gerald Combs, originally named **Ethereal**
- ▶ Renamed **Wireshark** in **2006** after a trademark dispute
- ▶ Built on **libpcap** (Linux / macOS) and **Npcap** (Windows)
- ▶ Ships with **3 000+** protocol dissectors out of the box
- ▶ Cross-platform GUI, plus the **tshark** command-line equivalent

Key takeaway — *Wireshark is older than most students in this room. Its dissectors are the de-facto specification of how the wire really looks.*



Two filter languages, one rule

- ▶ **Capture filters** use BPF — applied before packets are saved (lightweight)
- ▶ **Display filters** use Wireshark syntax — applied after capture (powerful)
- ▶ Save pcaps for offline forensic analysis and chain-of-custody
- ▶ Real-life: Toronto credit-union audit found a printer spewing **NTLMv1** auth
- ▶ Ethics: capture **only on networks you own or have written authorisation for**

Key takeaway — *Filter early to cut volume, filter late to refine analysis. Document both filters in your lab report.*



Network discovery and inventory

History and operation

- ▶ Released **1997** in Phrack magazine by Fyodor (Gordon Lyon)
- ▶ Sends crafted **TCP / UDP / ICMP** probes; interprets responses AND silences
- ▶ Open-source under the Nmap Public Source Licence (a custom GPL variant)
- ▶ **Zenmap** is the official cross-platform graphical front-end
- ▶ Powers asset inventory, vuln triage, network maps, compliance evidence

Key takeaway — *Nmap turns 'I don't know what is on my network' into a CSV inventory in twenty minutes.*



Flags you must memorise

- ▶ ``-sn`` ping sweep (host discovery, no port scan)
- ▶ ``-sS`` TCP SYN (stealth) scan — needs root / Administrator
- ▶ ``-sV`` service / version detection
- ▶ ``-O`` OS fingerprinting from TCP/IP idiosyncrasies
- ▶ ``--script`` NSE (Nmap Scripting Engine) — ~600 ready-made scripts
- ▶  Authorised scope only — see Canadian Criminal Code s.342.1 & US CFAA

Key takeaway — *Real-life Vancouver dental clinic: one ``nmap -sV`` revealed two unsupported Windows 7 hosts running EOL software. That scan saved a ransomware case.*



Time-series graphing for SMBs

- ▶ Released **2001** by Ian Berry, built on top of **RRDtool**
- ▶ Polls devices over **SNMP** every five minutes by default
- ▶ Stack: **PHP + MariaDB + Apache + cacti-spine poller**
- ▶ Pre-built templates for routers, switches, printers, UPSes, Linux servers
- ▶ Free, well-understood, runs on a Raspberry Pi if needed

Key takeaway — *Cacti is the 'graphs only' workhorse. It does one thing and does it cheap.*



Strengths and limits

- ▶ **Strength:** historical trends, capacity planning, 'what changed yesterday?'
- ▶ **Strength:** template community is huge — most vendors are covered
- ▶ **Limit:** read-only — no paging, no built-in alert engine
- ▶ **Limit:** 5-minute polling granularity hides fast spikes
- ▶ **Pair with Nagios** for alerts; **Grafana** for slicker dashboards

Key takeaway — *Cacti tells you what happened — not what to do about it. Pair it with Nagios.*



Availability monitoring

History and plugin model

- ▶ Created **1999** as **NetSaint** by Ethan Galstad — renamed Nagios 2002
- ▶ Open-source under the GPL; commercial **Nagios XI** tier exists
- ▶ **Plugin contract:** any program that exits 0/1/2/3 = OK/WARN/CRIT/UNKNOWN
- ▶ Anything you can script can become a check (Bash, Python, PowerShell)
- ▶ Web UI = red / yellow / green tiles per service

Key takeaway — *Nagios is brutally simple by design. Every check is a tiny script that exits with a number.*



Ecosystem and culture

- ▶ Same plugin API used by forks: **Icinga, Naemon, Shinken, Sensu**
- ▶ Heritage: pioneered the 'green dashboard or someone gets paged' culture
- ▶ Real-life Calgary law firm: Nagios paged the partner at 02:13 a.m. when their public Outlook Web Access dropped
- ▶ Modern alternatives layer on top: **Grafana OnCall, PagerDuty, Opsgenie**
- ▶ Learn the plugin contract once — it transfers to every fork

Key takeaway — *Pick any 'green-tiles' tool you like. Master the four-state exit contract and you can write checks for any of them.*



Simple Network Management Protocol

A 1988 protocol that runs every network on earth

- ▶ **v1** = RFC 1067 (1988) — clear-text community string
- ▶ **v2c** = 1996 — adds bulk operations, still clear-text
- ▶ **v3** = 1998 — authentication + encryption (SHA + AES)
- ▶ Despite its age, v1/v2c are still the default on most cheap switches
- ▶ OIDs are organised in **MIBs** (Management Information Bases)

Key takeaway — *If a device speaks SNMPv3, configure SNMPv3. If it only speaks v2c, ACL-lock the SNMP service to your monitor host only.*



Roles and operations

- ▶ **Agent** runs on every managed device (switch, router, printer, server)
- ▶ **Manager** = NMS (Network Management Station) that queries the agents
- ▶ **Operations:** `snmpget` (one OID) · `snmpwalk` (a sub-tree) · `snmpbulkget` (many) · `snmptrap` (push)
- ▶ **UDP/161** for polling · **UDP/162** for traps
- ▶ Each OID is an **Object Identifier** — a numeric address inside the MIB tree

Key takeaway — *The four verbs (get, walk, bulkget, trap) cover 95% of all real-world SNMP. Memorise them.*



OIDs you must know

- ▶ `1.3.6.1.2.1.1.5.0` → sysName (hostname)
- ▶ `1.3.6.1.2.1.1.3.0` → sysUpTime (reboot detection)
- ▶ `1.3.6.1.2.1.2.2.1.10` → ifInOctets (interface bandwidth)
- ▶ `1.3.6.1.4.1.9.9.91.1.1.1.4` → Cisco entSensorValue (temperature)
- ▶ Halifax college example: polling entSensorValue every two minutes caught a switch climbing past 55 °C during a heat-wave **before** it shut its ports down.

Key takeaway — *SNMPv1 and v2c send their community string in cleartext. Treat them like passwords on a postcard.*



The universal log protocol

History and RFCs

- ▶ Invented by **Eric Allman** in the **early 1980s** for Sendmail at UC Berkeley
- ▶ Standardised first by **RFC 3164 (1984/2001)** — the 'BSD format'
- ▶ Replaced by **RFC 5424 (2009)** — adds structured data, time-zones, IDs
- ▶ Spoken by virtually every router, firewall, switch, server, IoT gateway
- ▶ **rsyslog** and **syslog-ng** are the two dominant daemons on Linux today

Key takeaway — *If a device cannot send syslog in 2026, it is not a serious infrastructure device. Treat that as a buying criterion.*



Priority, facilities, ports

- ▶ **Priority = facility × 8 + severity**
- ▶ **Facilities** include kern, user, mail, daemon, auth, syslog, local0–local7
- ▶ **Severities** range from **0 Emergency** down to **7 Debug** (next slide)
- ▶ Default transport: **UDP/514** · TCP/514 · TLS-protected **TCP/6514**
- ▶ Quebec credit-union example: OSFI B-13 requires 7-year retention — that decision is a syslog architecture decision

Key takeaway — *UDP/514 is fine for the lab; production traffic that matters legally must go over TCP with TLS.*



Memorise this table

Level	Keyword	Meaning	Typical action
-----	-----	-----	-----
0	Emergency	System is unusable	Page on-call immediately
1	Alert	Action must be taken immediately	Page on-call immediately
2	Critical	Critical conditions	Open P1 ticket
3	Error	Error conditions	Open P2 ticket
4	Warning	Warning conditions	Track in daily review
5	Notice	Normal but significant	Note in change log
6	Informational	Informational	Index only
7	Debug	Debug-level	Discard or separate index

Key takeaway — Most production environments alert on 0–3, log 4–6, drop 7. Default Cisco IOS chatters at level 6; tune it.



Five more open-source heavyweights

- ▶ **Zeek** (formerly Bro, 1995, Vern Paxson, LBNL) — connection-log generator
- ▶ **Suricata** (2009, OISF) — multi-threaded IDS / IPS / NSM; Snort-rule compatible
- ▶ **ntopng** — flow visualiser with a pretty web UI
- ▶ **LibreNMS** — fork of Observium; modern, prettier alternative to Cacti
- ▶ **Prometheus + Grafana** — the modern cloud-native metrics stack

Key takeaway — Modern enterprise stacks pair Zeek for visibility, Suricata for signatures, Prometheus for metrics, ELK for logs.



SECTION B

Day 05 — Demo The Lab Environment



VirtualBox · GNS3 · AWS Academy

Strengths

- ▶ **VirtualBox** — free, offline, fast snapshots
- ▶ **GNS3** — real Cisco / Juniper / pfSense images
- ▶ **AWS Academy Learner Lab** — real cloud, CloudWatch, GuardDuty, IAM
- ▶ Together: end-to-end coverage from VM to cloud

Weaknesses

- ▶ **VirtualBox** — hard to emulate a router fabric
- ▶ **GNS3** — RAM-hungry, image licensing complexity
- ▶ **AWS Academy** — time-limited credits, no persistent storage between sessions
- ▶ Each one alone leaves a gap in your skill set

Key takeaway — *A working analyst rarely sits in only one of these. Your portfolio is stronger with all three.*



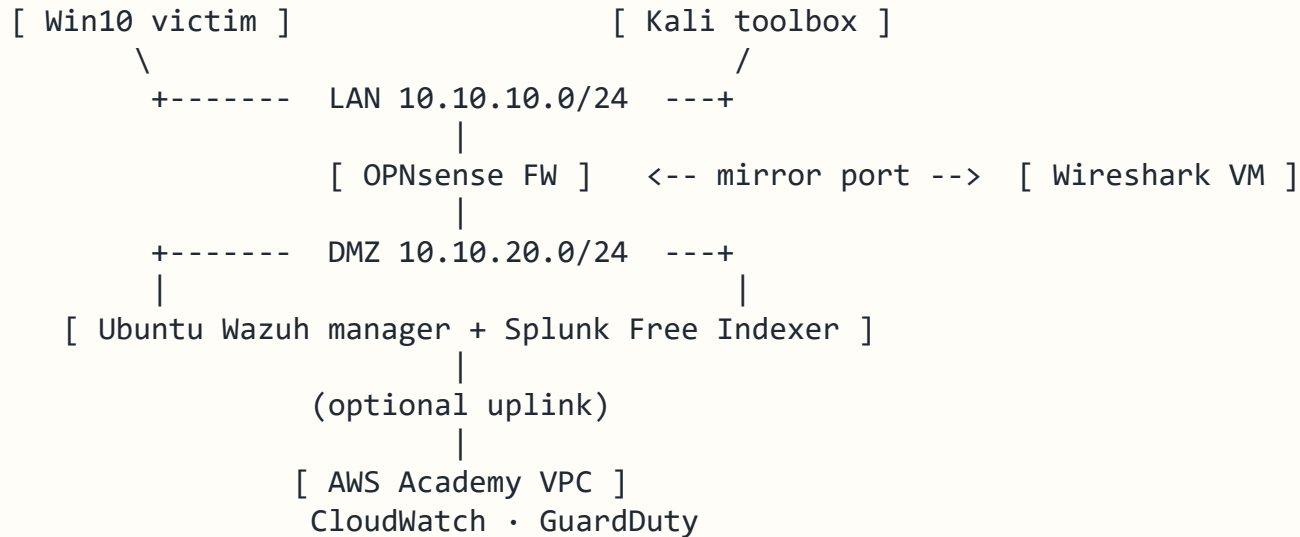
Other hypervisors worth knowing

- ▶ **VMware Workstation Pro** — free for personal use since May 2024
- ▶ **KVM + virt-manager** — Linux native; what most cloud providers run underneath
- ▶ **Proxmox VE** — Debian + KVM + LXC, free open-source hypervisor
- ▶ **EVE-NG** and **Containerlab** — GNS3 alternatives for network simulation
- ▶ **Cisco DevNet Sandbox** — free, hosted, reservable labs (CCNA / CCNP / DevNet)

Key takeaway — *Pick whichever tool you have time to master. Hiring managers care that you can build labs, not which hypervisor you used.*



What you will build by end of Day 05



Key takeaway — Host-only or NAT networking — never bridged. Snapshot every VM before any change.



SECTION C

Day 06 — Lecture 2

SIEM and Endpoint Detection



‘Google for log files’

History and licensing

- ▶ Founded **2003** in San Francisco — Erik Swan, Rob Das, Michael Baum
- ▶ Acquired by **Cisco in March 2024** for approximately US\$28 billion
- ▶ **Splunk Free**: 500 MB/day, no alerting, no clustering — perfect for class
- ▶ **Splunk Enterprise / Splunk Cloud**: paid by ingest GB or workload
- ▶ Real-life: Mont-Tremblant ski resort correlates chairlift turnstile + staff-door badge

Key takeaway — *Even at the free tier, Splunk teaches you the SIEM mindset — every keystroke counts when ingest is metered.*



Architecture & SPL

- ▶ Three roles: **Forwarder** · **Indexer** · **Search Head**
- ▶ In a lab, all three live on one VM; in production they scale out
- ▶ **SPL** (Search Processing Language) — pipe-delimited, like grep × awk on steroids
- ▶ Inputs: file, syslog UDP/514, **HEC** (HTTP Event Collector on TCP/8088)
- ▶ Custom dashboards, alerts, and saved searches form the SOC daily workflow

Key takeaway — *Learn SPL once and you can read any Splunk consultant's dashboard. It is the closest thing to a universal SIEM language.*



Open-source security platform

History and components

- ▶ Forked from **OSSEC** in **2015** by Santiago Bassett
- ▶ **Server** — rule engine, decoders, active-response orchestrator
- ▶ **Indexer** — runs OpenSearch under the hood
- ▶ **Dashboard** — Kibana-style web UI for analysts
- ▶ **Agents** — Windows / Linux / macOS / AIX / Solaris endpoints

Key takeaway — *Wazuh is a SIEM-lite + EDR-lite + vuln-scanner bundled in one open-source package. Hard to beat for the price.*



Features and integrations

- ▶ **HIDS** (Host Intrusion Detection) — log decoders + rules
- ▶ **FIM** (File Integrity Monitoring) — hash + attribute baselines
- ▶ **Vulnerability detection** — feed-driven CVE-to-package matching
- ▶ **MITRE ATT&CK** mapping per rule (T-numbers in every alert)
- ▶ Integrations: **VirusTotal, AbuseIPDB, Splunk, TheHive, MISP**
- ▶ Custom rules live in ``/var/ossec/etc/rules/local_rules.xml``

Key takeaway — *Toronto bookstore real-life: a USB-borne XMRig miner was caught by a 4-line custom Wazuh rule mapped to MITRE T1496.*



The log-search workhorse

History and the licence fork

- ▶ Elasticsearch released **2010** by Shay Banon
- ▶ Combined with **Logstash** and **Kibana** = the **ELK** stack (~2013)
- ▶ Elastic switched to **Elastic 2.0 / SSPL** in 2021 — no longer pure open-source
- ▶ AWS forked **OpenSearch + OpenSearch Dashboards** under **Apache 2.0** (2021)
- ▶ Wazuh runs on OpenSearch under the hood by default

Key takeaway — *If you only have time to learn one log backend deeply, learn OpenSearch — both Elastic and Wazuh shops will hire you.*



Typical pipeline

- ▶ **Beats / Vector / Fluent Bit** collect logs at the edge
- ▶ **Logstash** (or Vector) parses and enriches
- ▶ **OpenSearch** indexes and stores
- ▶ **OpenSearch Dashboards / Kibana** visualises and searches
- ▶ Production-scale: 100k+ events per second on a single tuned cluster

Key takeaway — *Beats → Logstash → OpenSearch → Kibana is the canonical pipeline. Memorise it; you will draw it in interviews.*



The pragmatic middle ground

- ▶ Created **2009** by Lennart Koopmann in Hamburg
- ▶ Same backend (OpenSearch / Elasticsearch) but a simpler operator experience
- ▶ **Streams** = pre-filtered views; **Content Packs** = importable rule bundles
- ▶ Free **Open** tier covers all SMB use cases
- ▶ Paid **Operations** / **Security** tiers add advanced correlation

Key takeaway — *Pick Graylog when you want SIEM simplicity over depth. It will not impress a Splunk veteran, but it will run for years untouched.*



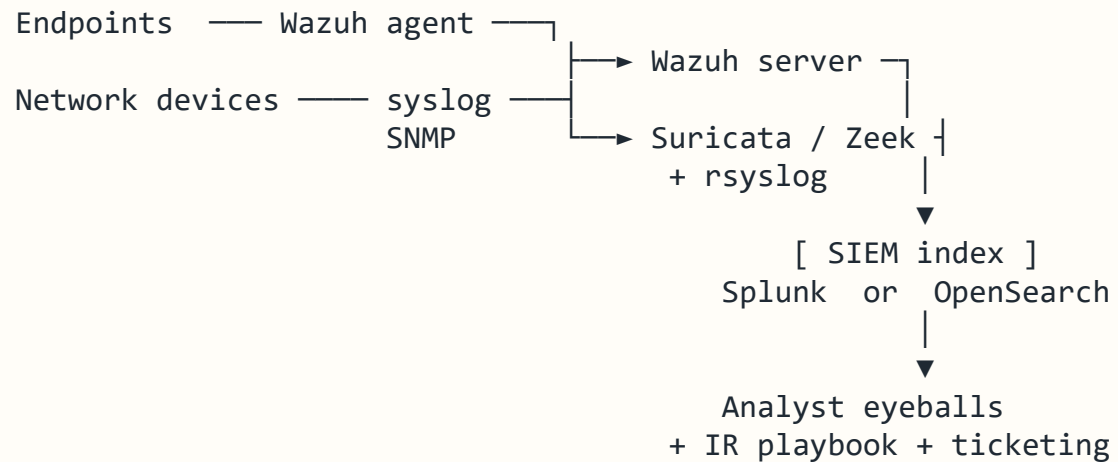
Drop-in SOC-in-a-box

- ▶ Started **2008** by Doug Burks as a single Ubuntu ISO
- ▶ Bundles **Zeek + Suricata + Wazuh + Elastic + CyberChef + TheHive**
- ▶ Aimed at education, MSP, and small-shop SOC use cases
- ▶ Free at the open tier; paid 'Security Onion Solutions' for support
- ▶ Great choice for the CYB-01 classroom virtual SOC

Key takeaway — *Pick Security Onion when you want a pre-integrated 'pull-and-pray' stack. Pick the individual tools when you need to customise.*



The mental model interviewers want



Key takeaway — Memorise this diagram — it is the answer to 'how would you monitor a 500-seat company?'



SECTION D

Day 06 — Hands-on Lab 3

Install • Configure • Verify



Infrastructure phases

- ▶ **Phase 1** — Bring up Day-05 topology (VirtualBox + GNS3 + OPNsense)
- ▶ **Phase 2** — Install Wireshark on the capture VM; configure WAN port mirror
- ▶ **Phase 3** — Enable SNMP on OPNsense and the lab switches; verify with `snmpwalk`
- ▶ Snapshot every VM before starting each phase
- ▶ Each phase has a checklist; you tick it off in your write-up

Key takeaway — *Phases 1–3 are infrastructure. They are graded on reproducibility, not creativity.*



Detection phases

- ▶ **Phase 4** — Install **Wazuh** single-node (manager + indexer + dashboard)
- ▶ **Phase 5** — Enrol a Windows 10 endpoint as a Wazuh agent
- ▶ **Phase 6** — Install **Splunk Free** and create a **HEC** (HTTP Event Collector) token
- ▶ **Phase 7** — Forward Wazuh alerts to Splunk; verify with one synthetic detection
- ▶ Optional: deploy the custom **XMRig rule** from ``tool10_wazuh_xmrig_rule.xml``

Key takeaway — *Phases 4–7 are detection. They are graded on whether the alerts actually appear, with screenshots as proof.*



Every chapter of your Course Procedure Manual must contain...

- ▶ **Title, author, date** (use the file-naming convention on slide 39)
- ▶ **Purpose** in five lines or fewer — answer 'why does this lab matter?'
- ▶ **Lab topology diagram** — draw.io export or hand-drawn scan are both fine
- ▶ **Step-by-step procedure** — numbered, with captioned screenshots
- ▶ **Verification** — show the result actually works (CLI output or screenshot)

Key takeaway — *A chapter is 'done' only when a friend who has never touched cyber can reproduce your lab from your document alone.*



Additional required sections

- ▶ **Troubleshooting** — at least three issues you hit, with their solutions
- ▶ **Reflection** — 200 words on what surprised you, what you would change
- ▶ **References** — every URL, every paper, every Stack Overflow link you used
- ▶ **Disclaimer block** — copy from the Day 05–06 lecture verbatim
- ▶ Export to **PDF**, NOT .docx, before upload — formatting must be locked

Key takeaway — *The troubleshooting and reflection sections are where most marks are won or lost. Do not skip them.*



Where, when, what

- ▶ **Where:** OmniVox → LEA → DropBox → '420-D01-AB Lab 3'
- ▶ **When:** Saturday 31 May, 23:59 EST
- ▶ **Naming:** `CYB01\_<surname>\_<firstname>\_Lab03.pdf`
- ▶ **Late policy:** –10% per calendar day (department rule, no exceptions)
- ▶ **Plagiarism:** all screenshots must be from YOUR lab; we check VM metadata

Key takeaway — *Twenty minutes of file-naming discipline saves an hour of TA chase-up. Use the exact format.*



The laws that apply to YOU

- ▶ **Canada** — Criminal Code s.342.1 (Unauthorized use of a computer)
- ▶ **United States** — Computer Fraud and Abuse Act, 18 U.S.C. § 1030
- ▶ **European Union** — Directive 2013/40/EU on attacks against information systems
- ▶ **Spain** — Ley Orgánica 10/1995, articles 197 bis & 264
- ▶ Scanning or sniffing outside your own lab subnet WILL get you expelled and possibly prosecuted

Key takeaway — *Cyber-defence tools become cyber-attack tools in unauthorised hands. The difference is consent in writing.*



Professional codes and habits

- ▶ **ACM Code of Ethics and Professional Conduct** — read it at acm.org/code-of-ethics
- ▶ **IEEE Code of Ethics** — read it at [ieee.org](https://www.ieee.org) → governance → P7-8
- ▶ Always paste suspicious URLs into **VirusTotal** BEFORE clicking
- ▶ Never reuse passwords between lab and personal accounts
- ▶ When in doubt, ask the instructor in writing — not Reddit, not Discord

Key takeaway — *Membership in the profession is a habit, not a certificate. The habits you build in this lab follow you into the SOC.*



Tonight's short list

- ▶ **Bejtlich, R.** *The Practice of Network Security Monitoring* (No Starch, 2013)
- ▶ **Chappell, L.** *Wireshark 101* (Laura Chappell University, 2017)
- ▶ **Clark, B. & White, A.** *Blue Team Field Manual (BTFM)* (CreateSpace, 2017)
- ▶ **Sanders, C.** *Practical Packet Analysis* (No Starch, 4th ed.)
- ▶ **NIST SP 800-94 Rev. 1** — Guide to Intrusion Detection and Prevention Systems

Key takeaway — *Tools age, principles don't. Read Bejtlich for the why, screenshot Wireshark for the how.*



QUESTIONS, NEXT STEPS, AND THANK YOU

Cyber.SoHo Educational Hub

From the Industry to the Classroom — Building the IT's Future, Today and Together!

Next class: Mid-Term Exam (Class 7) — covers Days 01 to 06.

Office hours and review session details posted on OmniVox.